Identifying equivalent hibernate 3 and hibernate 5 queries

Splunk query prefix to put to get DBTelemetry logs (which log the sql strings)

splunk_query_prefix

```
index="psp_iks" "DbTelemetry" "MapConsolidatorLogFlush"
```

1. Search for query comment if available

Search using the query comment. Note that hibernate 3 and hibernate 5 have minor differences when it comes to named queries. Hibernate 3 would add perfix such as 'named HQL query' before the queryname

1.1.1. named hql query difference

- hibernate 3 query comment - /* named HQL query findBillPaymentFinTxnsForOffloadBatch */
- hibernate 5 query comment - /* findBillPaymentFinTxnsForOffloadBatch */

1.1.2. named native sql query comment difference

- hibernate 3 query comment - /* named native SQL query findRangePartitions */
- hibernate 5 query comment - /* findRangePartitions */

1.1.3. hql query comment example (same query comment in both hibernate 3 and hibernate 5)

- query comment - /* select max(ee.SourceEmployeeId), min(ee.SourceEmployeeId) from com.intuit.sbd.payroll.psp.domain.Employee ee where ee.Company.SourceCompanyId = :companyId and ee.Company.SourceSystemCd = :systemCd */

1.2. Search the table names in from clause

Since most of the changes are in column aliases thus mostly this portion should remain same, If we are querying over a few tables, this should help to quickly identify the query

1.2.1. Example from clause

from_clause

```
from PSP_FINANCIAL_TRANSACTION d1_ inner join PSP_TRANSACTION_TYPE t0x1_ on d1_.TRANSACTION_TYPE_FK=t0x1_.
TRANSACTION_TYPE_CD inner join PSP_TRANSACTION_STATE t1x2_ on d1_.CURRENT_TRANSACTION_STATE_FK=t1x2_.
TRANSACTION_STATE_CD
```

1.2.2. hibernate 3 query

from_clause_hb3

```
/* criteria query */ select count(distinct this_.PAYROLL_RUN_SEQ) as y0_ from PSP_PAYROLL_RUN this_ where
(((this_.COMPANY_FK=:1 and not this_.PAYROLL_RUN_STATUS in (:2 )) and not exists (/* criteria query */ select
d1_.PAYCHECK_SEQ as y0_ from PSP_PAYCHECK d1_ where lower(d1_.SOURCE_PAYCHECK_ID) like :3 and d1_.
PAYROLL_RUN_FK=this_.PAYROLL_RUN_SEQ)) and ((not this_.PAYROLL_RUN_TYPE in (:4 ) or (this_.PAYROLL_RUN_TYPE in
(:5 ) and this_.PAYROLL_DIRECT_DEPOSIT_AMOUNT>:6 )) or exists (/* criteria query */ select d1_.
FINANCIAL_TRANSACTION_SEQ as y0_ from PSP_FINANCIAL_TRANSACTION d1_ inner join PSP_TRANSACTION_TYPE t0x1_ on
d1_.TRANSACTION_TYPE_FK=t0x1_.TRANSACTION_TYPE_CD inner join PSP_TRANSACTION_STATE t1x2_ on d1_.
CURRENT_TRANSACTION_STATE_FK=t1x2_.TRANSACTION_STATE_CD where ((((t0x1_.ASSOCIATION_TYPE=:7 or t0x1_.
TRANSACTION_TYPE_CD in (:8 , :9 )) and not t1x2_.TRANSACTION_STATE_CD in (:10 , :11 )) and d1_.
SETTLEMENT_TYPE_CD=:12 ) and d1_.FINANCIAL_TRANSACTION_AMOUNT>:13 ) and d1_.PAYROLL_RUN_FK=this_.
PAYROLL_RUN_SEQ)))
```

1.2.3. hibernate 5 query

from_clause_hb5

```
/* criteria query */ select count(distinct this_.PAYROLL_RUN_SEQ) as y0_ from PSP_PAYROLL_RUN this_ where
(((this_.COMPANY_FK=? and not (this_.PAYROLL_RUN_STATUS in (?))) and not exists (/* criteria query */ select
d1_.PAYCHECK_SEQ as y0_ from PSP_PAYCHECK d1_ where lower(d1_.SOURCE_PAYCHECK_ID) like ? and d1_.
PAYROLL_RUN_FK=this_.PAYROLL_RUN_SEQ)) and ((not (this_.PAYROLL_RUN_TYPE in (?)) or (this_.PAYROLL_RUN_TYPE in
(?) and this_.PAYROLL_DIRECT_DEPOSIT_AMOUNT>?)) or exists (/* criteria query */ select d1_.
FINANCIAL_TRANSACTION_SEQ as y0_ from PSP_FINANCIAL_TRANSACTION d1_ inner join PSP_TRANSACTION_TYPE t0x1_ on
d1_.TRANSACTION_TYPE_FK=t0x1_.TRANSACTION_TYPE_CD inner join PSP_TRANSACTION_STATE t1x2_ on d1_.
CURRENT_TRANSACTION_STATE_FK=t1x2_.TRANSACTION_STATE_CD where ((((t0x1_.ASSOCIATION_TYPE=? or t0x1_.
TRANSACTION_TYPE_CD in (?, ?)) and not (t1x2_.TRANSACTION_STATE_CD in (?, ?))) and d1_.SETTLEMENT_TYPE_CD=?)
and d1_.FINANCIAL_TRANSACTION_AMOUNT>?) and d1_.PAYROLL_RUN_FK=this_.PAYROLL_RUN_SEQ)))
```

1.3. Add the where clause portion as well

If we are getting too many queries till step 2 (might be due to many similar queries or might be because very few tables are being queried) start adding portions in where clause and further

Note that we are querying just one table in following query, so it might be difficult to identity it just using step 2

1.3.1. Example search query

where_clause_example_query

```
from PSP_COMPANY_EVENT this_ where ((this_.COMPANY_FK=? and this_.EVENT_TIME_STAMP>=
? ) and this_.EVENT_TIME_STAMP<=? )
```

1.3.2. hibernate 3 query

where_clause_hb3

```
select * from ( /* criteria query */ select this_.COMPANY_EVENT_SEQ as COMPANY1_32_0_, this_.VERSION as
VERSION32_0_, this_.CREATOR_ID as CREATOR3_32_0_, this_.CREATED_DATE as CREATED4_32_0_, this_.MODIFIER_ID as
MODIFIER5_32_0_, this_.MODIFIED_DATE as MODIFIED6_32_0_, this_.REALM_ID as REALM7_32_0_, this_.EVENT_TIME_STAMP
as EVENT8_32_0_, this_.STATUS_EFFECTIVE_DATE as STATUS9_32_0_, this_.STATUS_CD as STATUS10_32_0_, this_.
EVENT_TYPE_CD as EVENT11_32_0_, this_.EVENT_TOKEN as EVENT12_32_0_, this_.SOURCE_ID as SOURCE13_32_0_, this_.
NOTE_LAST_UPDATED_DATE as NOTE14_32_0_, this_.COMPANY_FK as COMPANY15_32_0_ from PSP_COMPANY_EVENT this_ where
((this_.COMPANY_FK=? and this_.EVENT_TIME_STAMP>=
? ) and this_.EVENT_TIME_STAMP<=? ) order by this_.EVENT_TIME_STAMP desc, this_.EVENT_TYPE_CD asc, this_.
COMPANY_EVENT_SEQ asc ) where rownum <= ?
```

1.3.3. hibernate 5 query

where_clause_hb5

```
/* criteria query */ select this_.COMPANY_EVENT_SEQ as company_event_seq1_55_0_, this_.VERSION as
version2_55_0_, this_.CREATOR_ID as creator_id3_55_0_, this_.CREATED_DATE as created_date4_55_0_, this_.
MODIFIER_ID as modifier_id5_55_0_, this_.MODIFIED_DATE as modified_date6_55_0_, this_.REALM_ID as
realm_id7_55_0_, this_.EVENT_TIME_STAMP as event_time_stamp8_55_0_, this_.STATUS_EFFECTIVE_DATE as
status_effective_d9_55_0_, this_.STATUS_CD as status_cd10_55_0_, this_.EVENT_TYPE_CD as event_type_cd11_55_0_,
this_.EVENT_TOKEN as event_token12_55_0_, this_.SOURCE_ID as source_id13_55_0_, this_.NOTE_LAST_UPDATED_DATE as
note_last_updated14_55_0_, this_.COMPANY_FK as company_fk15_55_0_ from PSP_COMPANY_EVENT this_ where ((this_.
COMPANY_FK=? and this_.EVENT_TIME_STAMP>=?) and this_.EVENT_TIME_STAMP<=?) order by this_.EVENT_TIME_STAMP
desc, this_.EVENT_TYPE_CD asc, this_.COMPANY_EVENT_SEQ asc fetch first ? rows only
```

1.4. Beware of the cross join

We have seen that at many places hibernate 5 uses 'cross join' where hibernate 3 used to have implicit join syntax ','

While searching using the table names just make sure this this is not the cause of an issue.

Based on what we have seen this mostly happens only in cases of hql queries which we should be able to figure out using step 1 itself but still logging in this document

1.4.1. hibernate 3 query

cross_join_hb3

```
/* named HQL query findDailyAdjustments */ select sum(liabilitya0_.AMOUNT) as col_0_0_, sum(liabilitya0_.
TAXABLE_WAGES) as col_1_0_, liabilitya0_.LAW_FK as col_2_0_, liabilitya0_.EFFECTIVE_DATE as col_3_0_ from
PSP_LIABILITY_ADJUSTMENT liabilitya0_, PSP_LAW law1_, PSP_PAYMENT_TEMPLATE paymenttem2_, PSP_PAYROLL_RUN
payrollrun3_ where liabilitya0_.LAW_FK=law1_.LAW_ID and law1_.PAYMENT_TEMPLATE_FK=paymenttem2_.
PAYMENT_TEMPLATE_CD and liabilitya0_.PAYROLL_RUN_FK=payrollrun3_.PAYROLL_RUN_SEQ and liabilitya0_.
EFFECTIVE_DATE>=paymenttem2_.SUPPORT_START_DATE and liabilitya0_.IS_RECONCILING_ADJUSTMENT=0 and (payrollrun3_.
PAYCHECK_SETTLEMENT_DATE<=? or payrollrun3_.PAYROLL_RUN_STATUS='Complete') and payrollrun3_.
PAYROLL_RUN_STATUS<>'Pending' and liabilitya0_.EFFECTIVE_DATE>=? and liabilitya0_.EFFECTIVE_DATE<=? and
payrollrun3_.COMPANY_FK=? and payrollrun3_.PAYROLL_RUN_STATUS<>'Superseded' and  not (exists (select 'T' from
PSP_PAYCHECK paycheck8_ where paycheck8_.PAYROLL_RUN_FK=liabilitya0_.PAYROLL_RUN_FK and (paycheck8_.
SOURCE_PAYCHECK_ID like '-%'))) group by liabilitya0_.LAW_FK , liabilitya0_.EFFECTIVE_DATE
```

1.4.2. hibernate 5 query

cross_join_hb3

```
/* findDailyAdjustments */ select sum(liabilitya0_.AMOUNT) as col_0_0_, sum(liabilitya0_.TAXABLE_WAGES) as
col_1_0_, liabilitya0_.LAW_FK as col_2_0_, liabilitya0_.EFFECTIVE_DATE as col_3_0_ from
PSP_LIABILITY_ADJUSTMENT liabilitya0_ cross join PSP_LAW law1_ cross join PSP_PAYMENT_TEMPLATE paymenttem2_
cross join PSP_PAYROLL_RUN payrollrun3_ where liabilitya0_.LAW_FK=law1_.LAW_ID and law1_.
PAYMENT_TEMPLATE_FK=paymenttem2_.PAYMENT_TEMPLATE_CD and liabilitya0_.PAYROLL_RUN_FK=payrollrun3_.
PAYROLL_RUN_SEQ and liabilitya0_.EFFECTIVE_DATE>=paymenttem2_.SUPPORT_START_DATE and liabilitya0_.
IS_RECONCILING_ADJUSTMENT=0 and (payrollrun3_.PAYCHECK_SETTLEMENT_DATE<=? or payrollrun3_.
PAYROLL_RUN_STATUS='Complete') and payrollrun3_.PAYROLL_RUN_STATUS<>'Pending' and liabilitya0_.
EFFECTIVE_DATE>=? and liabilitya0_.EFFECTIVE_DATE<=? and payrollrun3_.COMPANY_FK=? and payrollrun3_.
PAYROLL_RUN_STATUS<>'Superseded' and  not (exists (select 'T' from PSP_PAYCHECK paycheck8_ where paycheck8_.
PAYROLL_RUN_FK=liabilitya0_.PAYROLL_RUN_FK and (paycheck8_.SOURCE_PAYCHECK_ID like '-%'))) group by
liabilitya0_.LAW_FK , liabilitya0_.EFFECTIVE_DATE
```